Theme 3 – Challenges in the Data Economy – Ethical perspectives on data-driven management

From analytics to action

Follow-up question Rigshospitalet

Today

- Learning objective #1
- Analyze ethical challenges and opportunities in a specific data analysis project.

- How do you analyze ethical challenges? Can they be an opportunity at the same time?

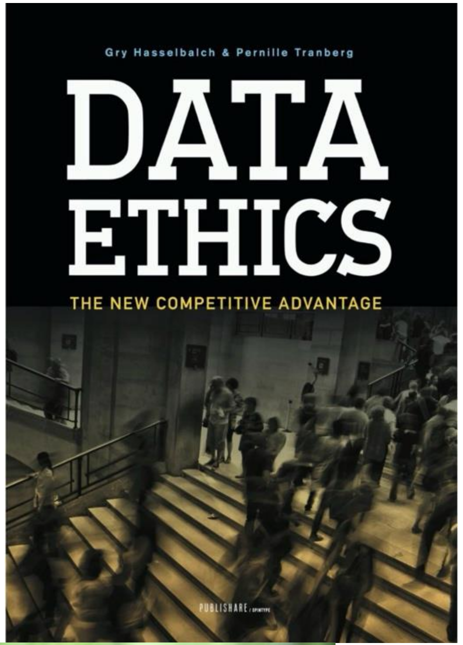
- Start with thinking about **power relations**: Who has the power to raise issues, define problems, or propose and create the solutions in relation to data-driven management? How could you conduct a data interest analysis for an organization (but also beyond)?

1. Concepts:
   - Data ethics (of power)
   - Data interest analysis
   - Ethics as data innovation
   - 4 models of data governance
     1. data sharing pools
     2. data cooperatives
     3. public data trusts
     4. personal data sovereignty

Teaching you two things at a time

- How to analyze challenges in the data economy **using the concept of data ethics**
- Understanding the use of social science concepts in relation to an organizational problem

Data ethics: the new competitive advantage (Hasselbalch and Tranberg, 2016: 9)



"We are living in an era defined and shaped by data. Data makes the world go round. It is politics, it is culture, it is everyday life and it is business. Our data-flooded era is one of technological progress, with tides rising at a never seen before pace. Roles, rights and responsibilities are reorganised and new ethical questions posed. Data ethics must and will be a new compass to guide us."

Data ethics failure 1: Facebook – Cambridge Analytica data scandal (2014)

- **The issue:**
  - Data from about 87 million Facebook users was collected through a personality quiz app
  - Most users did not consent to their data being used

- **How the data was used**
  - To create psychological profiles of voters
  - Used for targeted political advertising

- **Ethical issues**
  - Lack of informed consent
  - Secondary use of personal data
  - Potential manipulation of democratic processes

Data ethics failure 2: Dieselgate - Volkwagen (2015)

- **The issue:**
  - Volkswagen installed software in diesel cars that detected emissions testing conditions
- **How the data was used:**
  - The car's system used sensor data to recognize test environments
  - It changed engine behaviour to pass tests. During normal driving, emissions were far higher than reported
- **Ethical issues:**
  - Intentional algorithmic deception
  - False environmental data
  - Harm to public health and trust

Data ethics failure 3: UBER's 'God view' controversy (2014)

- **This issue:**
  - Uber employees had access to a tool called “God View”
  - It allowed them to see real-time locations of riders and drivers

- **Ethical problems**
  - Employees reportedly tracked journalists and customers
  - Users were not aware this level of monitoring existed

- **Ethical issues**
  - Privacy violations
  - Abuse of internal data access
  - Lack of data governance

Uber settles 'God View' allegations

KAJA WHITEHOUSE | USA TODAY

NEW YORK — Ride-hailing giant Uber has agreed to encrypt rider geo-location to settle a case tied to allegations that the company allowed its executives to track riders' whereabouts without their permission.

Uber's agreement with New York Attorney General Eric Schneiderman closes an investigation sparked by reports that its executives had access to riders' locations, and that Uber displayed rider information in an aerial view, known internally as “God View.”

Travis Kalanick, Founder and CEO of Uber, delivers a speech at the Institute of Directors Convention at the Royal Albert Hall, Central London, Britain, on ... **Show more**

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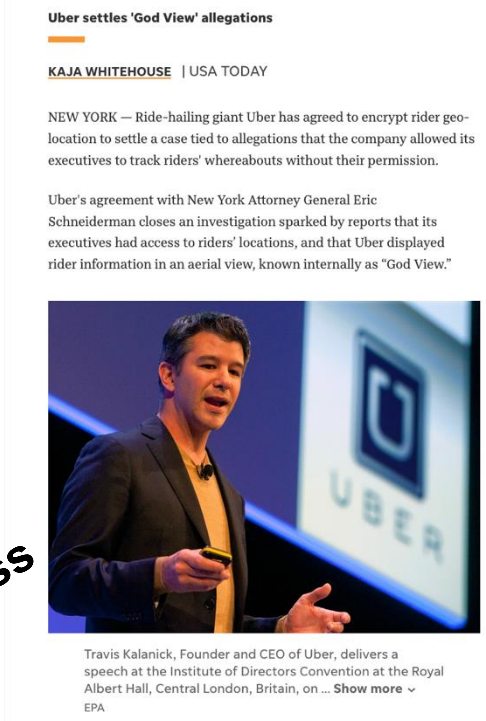
What categories of unethical data use do these 3 cases represent?

misuse of personal data

Manipulation of data systems



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Abuse of data access

What is an ethical company in the Big Data era? (Hasselbalch and Tranberg, 2016: 10-11)

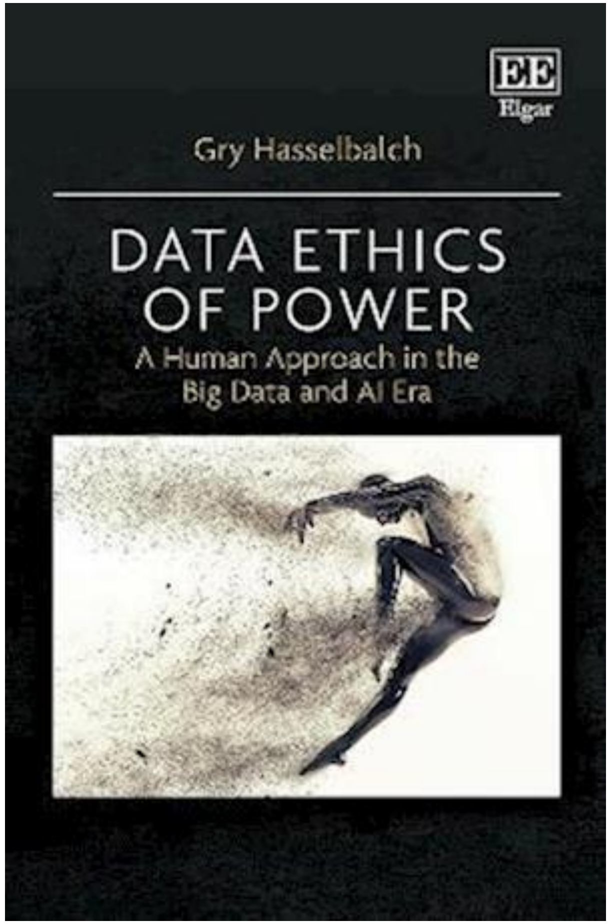
- **Go beyond legal compliance** with data protection laws
- **Follow the spirit of legislation**, not just the rules
- **Actively listen to customers** about how their data is used
- **Maintain clear and credible transparency** in data management
- **Collect and process only necessary data** (data minimisation)
- **Build a privacy-aware corporate culture and organisational structure**
- **Develop products using *Privacy by Design***
- **Use ethical reflection in decisions**, asking:
  - *Would I accept this as a consumer?*
  - *Is this the kind of data environment I want for my children?*

What is "Data Ethics of Power" (Hasselbalch, 2021)

"Data ethics (of power) addresses the distribution of power and power relations in the Big Data Society and the conditions of their negotiation and distribution. Applied data ethics is concerned with making these power relations visible in order to point to design, business, policy, social and cultural processes that support a human(-centric) distribution of power."

(Hasselbalch, 2021: 10)

- Power dynamics and interests by design
- Whose interests are prioritized?
- Who is empowered? Who is disempowered?



Exercise about data ethics of power – smart city planning

Discuss in small groups the **ethical challenges in smart city projects** (e.g., surveillance, lack of citizen participation, etc.) and propose two ethical rules that cities should consider

- *Topics to discuss:*
  - What data should the city collect?
  - Who should own and control the data?
  - How should citizens be involved in decisions about data?
  - What rules should exist about how data is used, to prevent misuse, to ensure public oversight?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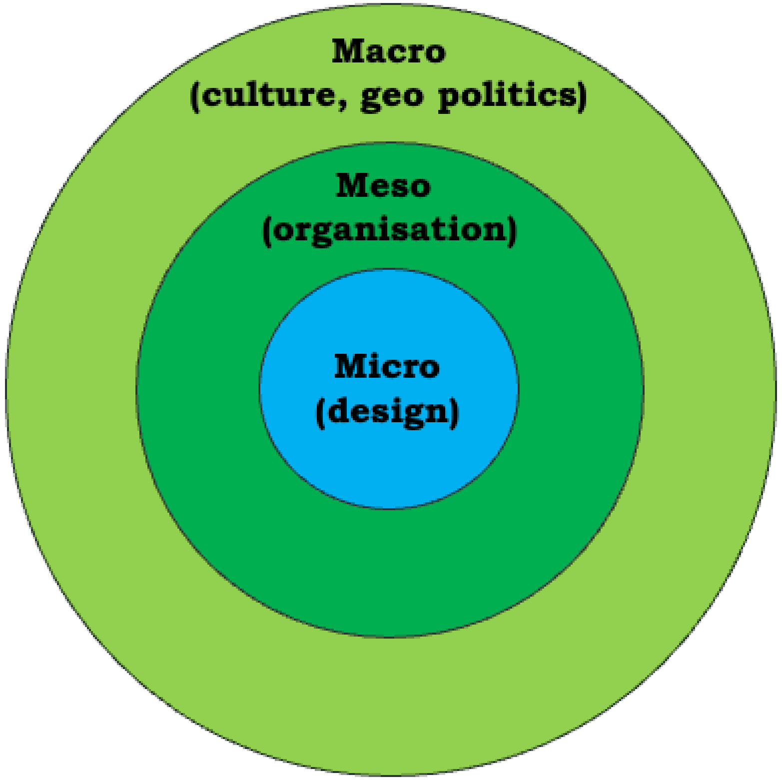
How to conduct a data interest analysis

- Multi-level analysis: examples on micro, meso, macro
- How different interests in data are empowered or disempowered by design



Analytical framework for infrastructures of power from *Data Ethics of Power*, p. 8-10

human agency and oversight

e.g. Is the AI system designed to interact, guide or take decisions by human end-users that affect humans or society?

technical robustness and safety

e.g. Could the AI system have adversarial, critical or damaging effects (e.g. to human or societal safety) in case of risks or threats such as design or technical faults, defects, outages, attacks, misuse, inappropriate or malicious use?

privacy and data governance

e.g. Did you consider the impact of the AI system on the right to privacy, the right to physical, mental and/or moral integrity and the right to data protection?

transparency

e.g. Did you put in place measures that address the traceability of the AI system during its entire lifecycle?

diversity, non-discrimination and fairness

e.g. Did you assess whether the AI system's user interface is usable by those with special needs or disabilities or those at risk of exclusion?

environmental and societal well-being

e.g. Are there potential negative impacts of the AI system on the environment?

accountability

e.g. Did you foresee any kind of external guidance or third-party auditing processes to oversee ethical concerns and accountability measures?

Any one of these ethical requirements could be a design choice (and a form of innovation)

– examples?

Ethics as data innovation

NHS signs up for Tim Berners-Lee pilot to reinvent web

BBC, NatWest and government of Flanders also test new Solid servers from Inrupt

Solid   About Solid   For Users   For Develop

Solid: Your data, your choice.

Advancing the Web to empower people and communiti

Read about Solid

Solid is an evolution of the web by its creator

Use Solid

BBC

Home News Sport Business Innovation Culture Arts Travel Earth Audio Video Live

Sir Tim Berners-Lee says individuals should claim back control of their personal data

"From the beginning, I always meant for the web to be a platform for creativity and collaboration," says Sir Tim Berners-Lee, inventor of the world wide web.

Could Sir Tim Berners-Lee one day unite Europe on a shared data platform?

Solid is a framework that enables individuals to store their personal data in “pods” (Personal Online Data Stores), giving them control over who accesses their information.

It represents a shift from centralized data exploitation to a user-centric, ethically designed data service.

1. User Data Ownership – Instead of companies storing user data on their servers, users keep their data in self-controlled pods.
2. Consent-Driven Access – Users explicitly decide which applications or companies can access specific data.
3. Decentralized Storage – Reduces risks of mass surveillance, data breaches, and exploitation by central authorities.
4. Interoperability & Transparency – Solid is open-source and built on semantic web standards

- Flanders, Belgium: The Flemish Government is integrating Solid for citizen-controlled digital identity and data sharing across government services.

MyData

About Participate Publications Events News

We help people and organisations to benefit from personal data in a human-centric way. To create a fair, sustainable, and prosperous digital society for all.

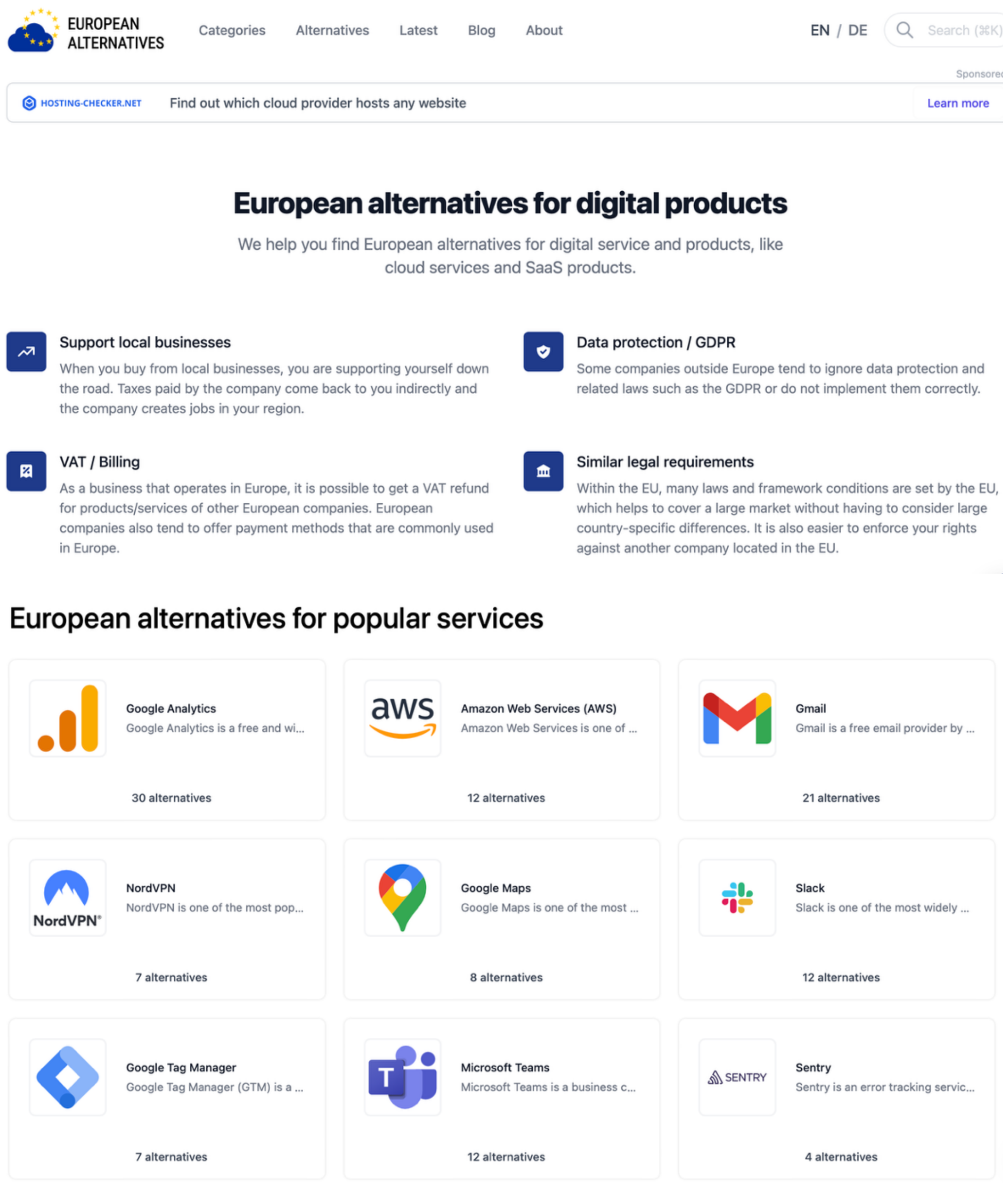
- We open space for **practical collaboration** through events and working groups.
- We **make sense** of rapid changes in the data environment.
- We contribute to the **emerging infrastructure** of data spaces and intermediaries.
- We help **set the agenda** for policymakers and changemakers.
- We promote **solutions and services** that put people first.

Become a Member | Sign the declaration | Read our Brochure

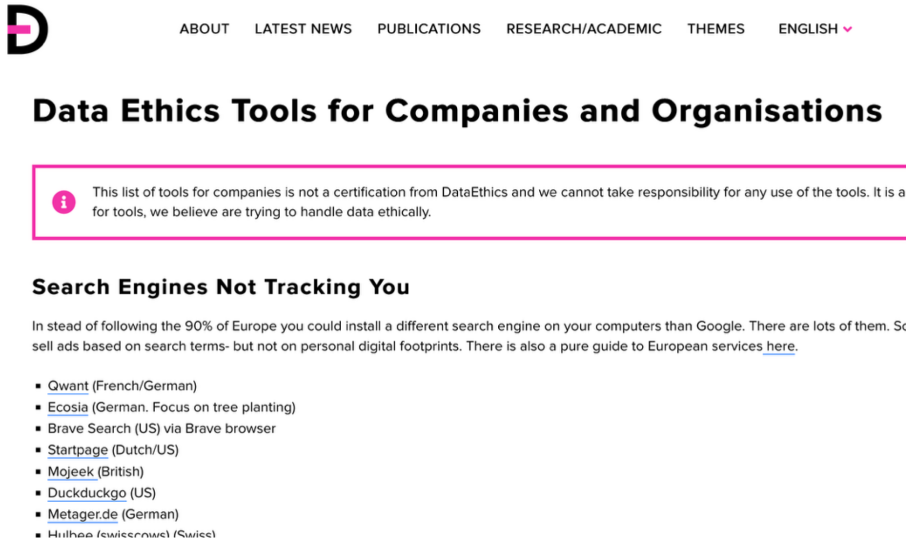
1740 signatories and counting

Who we are

- International nonprofit association with over **460 members** from **45 countries**
- Founded in 2018 and headquartered in Finland
- Local hubs on 6 continents and thematic groups developing standards and technical reference models
- 8-person secretariat and planned 914K€ annual budget
- Prominent position in multiple EU projects, including the Data Spaces Support Centre
- Organiser of the leading global conference on personal data



https://european-alternatives.eu/



https://dataethics.eu/

4 models of data governance (Micheli et al., 2020: 6)

Model	Who controls the data?	Main goal	Example
Data Sharing Pools (DSP)	Multiple organizations share and combine datasets	Innovation and economic value through combining data from different sources	Companies sharing mobility data for traffic services
Data Cooperatives (DC)	Citizens collectively manage and decide how their data is used	Empower citizens and create public or social value	Health data cooperatives like MIDATA
Public Data Trusts (PDT)	Public institutions manage data on behalf of citizens	Use data responsibly for policy making and public good	City governments managing urban data for planning
Personal Data Sovereignty (PDS)	Individuals control access to their own data	Data self-determination and user control	MyData movement personal data spaces

4 models of data governance (Micheli et al., 2020: 6)

Table 2. Summary of data governance models.

Model	Key actors	Goals	Value	Mechanisms
Data sharing pools (DSPs)	• Business entities • Public bodies	• Fill knowledge gaps through data sharing • Innovate and develop new services	• Private profit • Economic growth	• Principle of 'data as a commodity' • Partnerships • Contracts (e.g. repeatable frameworks)
Data cooperatives (DCs)	• Civic organisations • Data subjects	• Rebalance power unbalances of the current data economy • Address societal challenges • Foster social justice and fairer conditions for value production	• Public interest • Scientific research • Empowered data subjects	• Principles from the cooperative movement • Data commons • 'Bottom-up' data trusts • GDPR Right to data portability
Public data trusts (PDTs)	• Public bodies	• Inform policy-making • Address societal challenges • Innovate • Adopt a responsible approach to data	• Public interest • More efficient public service delivery	• Principle of 'data as a public infrastructure' • Trust building initiatives • Trusted intermediaries • Enabling legal framework
Personal data sovereignty (PDS)	• Business entities • Data subjects	• Data subjects self-determination • Rebalance power unbalances of the current data economy • Develop new digital services • centred on users need	• Empowered data subjects • Economic growth • Private profit • Knowledge	• Principle of 'technological sovereignty' • Communities and movements (e.g. MyData) • Intermediary digital services (personal data spaces) • GDPR Right to data portability

Example for data sharing pools

Healthcare Data Sharing Pool: Hospitals in a town seeking to improve predictive models for disease outbreaks without violating patient privacy

- Each hospital contributes anonymized patient records (like age, symptoms, diagnoses) to a shared data pool
- Data is stored in a secure, centralized or federated system that ensures no hospital can access another hospital's raw data
- Researchers or AI systems can query the pooled data to detect trends, train machine learning models, or perform statistical studies
- Access is governed by strict policies, often monitored to comply with laws

Result: Hospitals benefit from **better disease prediction models** that wouldn't be possible with their own data alone. **Patient privacy** is maintained. Innovation and **insights are shared across the healthcare system** efficiently.

Example for data cooperative

Personal Mobility Data Cooperative: Citizens in a city want to improve urban transportation planning without giving up control of their personal travel data.

- Individuals contribute anonymized data from their smartphones or fitness trackers (like walking routes, cycling paths, or commuting patterns)
- The data is pooled in a **cooperative platform** where members collectively govern access policies - deciding which city planners, researchers, or apps can use it
- Any insights, such as suggestions for new bike lanes or traffic flow optimizations, are shared back with the cooperative members or the city
- Members may also receive rewards, discounts, or improved services in return for contributing their data

Result: Citizens retain control over their data instead of handing it to a single company. **Urban planners get rich, aggregated insights** to improve city infrastructure**. Benefits are shared fairly among cooperative members** rather than centralized in a private company.

Example for public data trust

Environmental Data Public Trust: A country wants to monitor and improve environmental health, like air quality, water pollution, and biodiversity, without letting private companies control the data.

- Government agencies, research institutes, and private organizations contribute environmental data (e.g., pollution readings, satellite imagery)
- A public data trust is established, with a legal trustee responsible for managing access and ensuring compliance with privacy, ethics, and environmental regulations
- Researchers, NGOs, and policymakers can access the data under strict rules for public-benefit projects
- The trust enforces that any commercial use of the data must align with public interest (e.g., funding conservation projects)

Result: Data is used transparently and ethically for environmental protection. Public interest is safeguarded, no single entity owns or misuses the data. The community benefits from better research, policy decisions, and environmental outcomes.

Example for personal data sovereignty

- **Smart Home Data Sovereignty:** A person owns smart home devices such as thermostats, security cameras, and voice assistants, but doesn’t want the companies to collect or sell their data without control
  – All device data (e.g., temperature preferences, camera footage, voice commands) is stored in a personal data vault controlled by the homeowner
  – The individual decides: Which apps or services can access the data, for how long they can use it (e.g., only during troubleshooting), whether the data can be anonymised and shared for research, smart home improvement, or energy efficiency studies

Access can be revoked anytime, and the homeowner can monitor exactly who has used their data

Results: The individual retains ownership and control over sensitive daily-life data. Smart home companies can only use data if explicitly permitted, promoting privacy. The user can gain benefits (like better energy recommendations) **without surrendering their data rights**.

Mid-term evaluation

Thank you for taking time to fill in the mid-term evaluation!

DTU